Cap 12 - Fções Booleanas

Exemplos

Ex 12.1 Apenas explicativo

Ex 12.2 $f(x,y,z) = xyz' + xy'z + xy'z' + x'yz + x'y'z'$

$$xy'\underbrace{(z+z')}_{1} + x'z'\underbrace{(y+y')}_{1} + xyz'$$

$$xy' + x'z' + xyz'$$

$$x'z' + x(y' + yz')$$

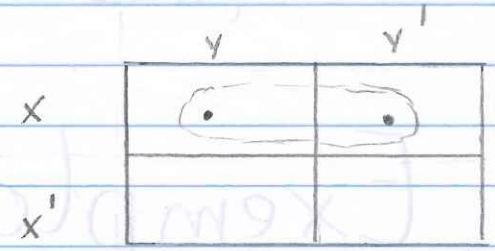
$$x'z' + x\underbrace{((y'+y)}_{1}.(y'+z'))$$

$$x'z' + xy' + xz'$$

$$z'\underbrace{(x+x')}_{1} + xy'$$

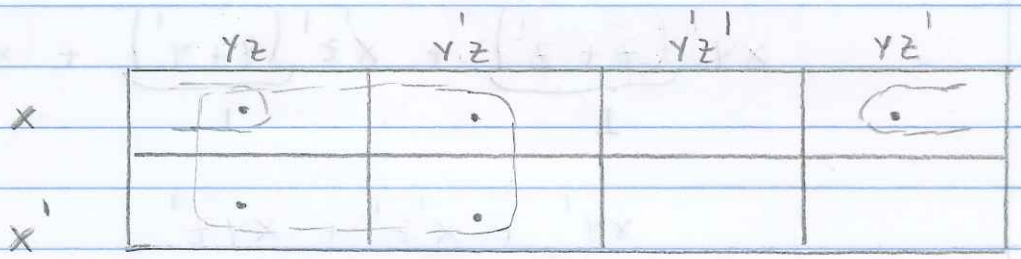
$$\boxed{z' + xy'}$$

Ex 12.3 a) $f(x,y) = xy + xy'$



$$f(x) = x$$

b) $f(x,y,z) = xyz + xyz' + xy'z + xy'z' + x'yz + x'y'z$



$$f(x,y,z) = z + xy$$

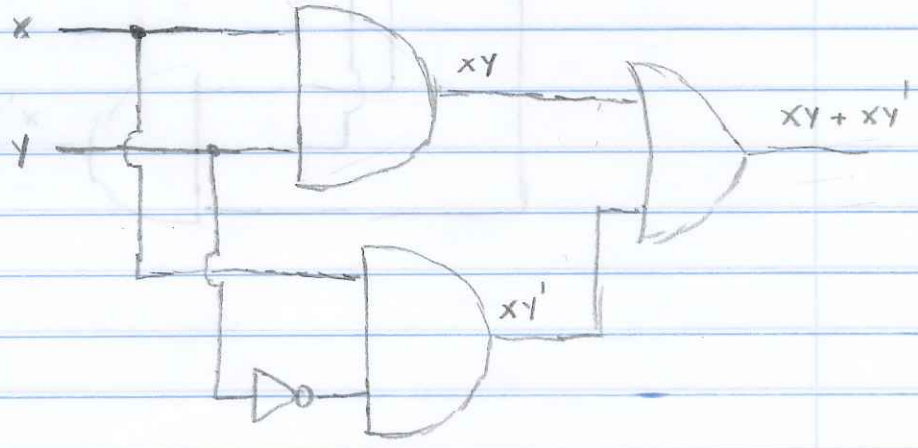
c) $f(x, y, z, t) = xy' + xyz + x'y'z' + x'yzt'$

	zt	zt'	$z't'$	$z't$
xy	•	•		
xy'	•	•	•	•
$x'y'$			•	•
$x'y$		•		

$$f(x, y, z, t) = xz + y'z' + yzt'$$

Ex 12.4

a) Função original ⇒ $f(x, y) = xy + xy'$



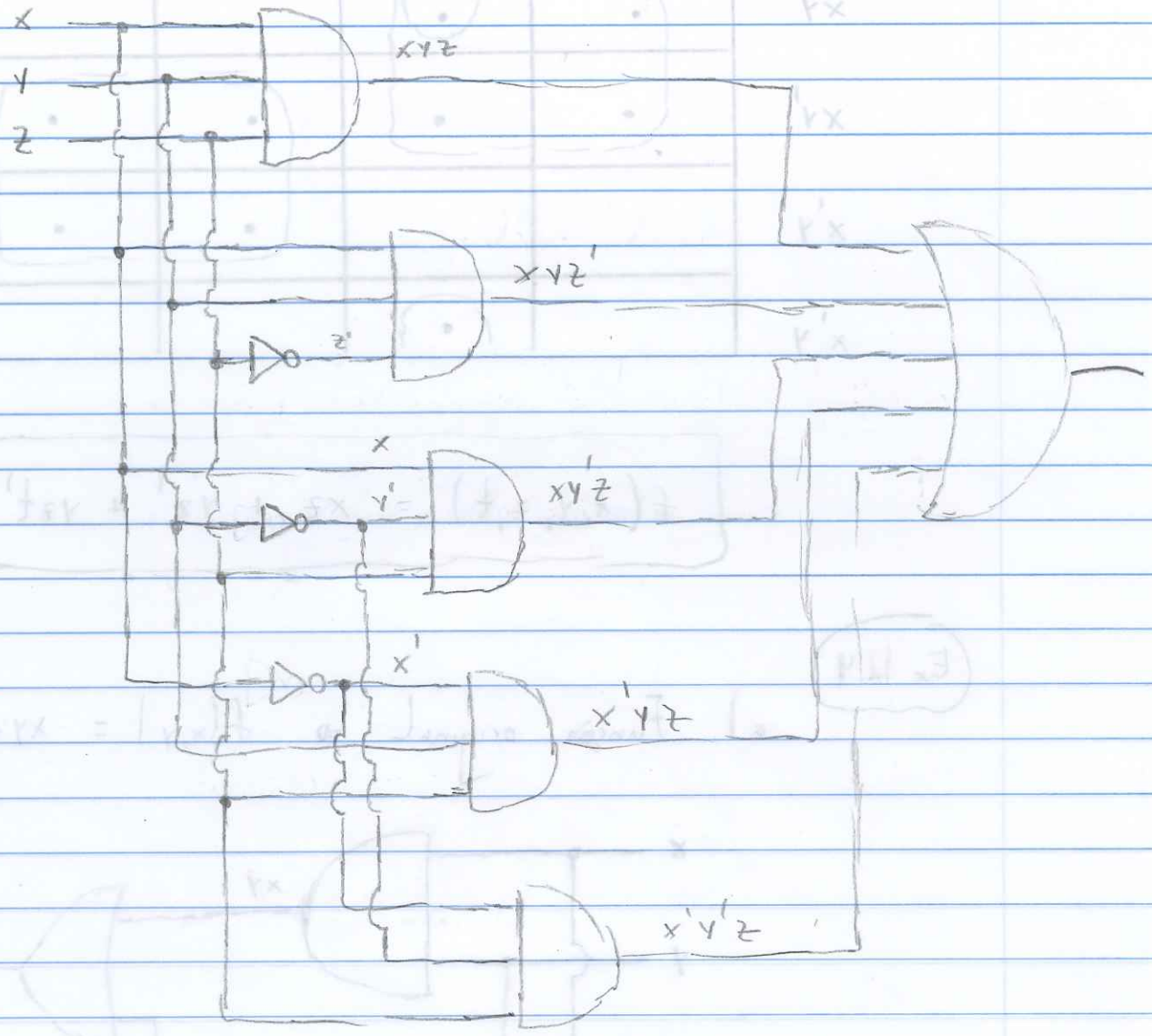
Função simplificada ⇒ $f(x) = x$

x ——— x

(04)

b) Original:

$$f(x,y,z) = xyz + xyz' + xy'z + x'yz + x'y'z$$



Lista 12 - Álgebra Booleana

1 a) $f(x,y,z) = xyz + xz' + xy'$

$= x(yz + z' + y')$

$= x\underbrace{[yz + (yz)']}_{1}$ MD

$$\boxed{f(x,y,z) = x}$$

b) $f(x,y,z) = (xyz')(x' + y' + z')$

$= \underbrace{xyzx'}_{0} + \underbrace{xyz'y'}_{0} + \underbrace{xyz'z'}_{xyz'}$ MD

$$\boxed{f(x,y,z) = xyz'}$$

c) $f(x,y,z,t) = ((xz)' + y + t)' + z(x + z + t)'$

$(x' + z' + y + t)' + z(x' + z' + t')$

$(x' + z' + y + t)' + zx' + \underbrace{zz'}_{0} + zt'$

$xzy't' + x'z + zt'$

$x'z + zt'\underbrace{(xy' + 1)}_{1}$

$x'z + zt' = z(x' + t')$

$$\boxed{f(x,y,z,t) = z(x' + t')}$$

01

d) $f(x,y,z,t) = [(x+y).z]' + [t(z+y)]'$

$(x+y)' + z' + t' + (z+y)'$

$x'y' + z' + t' + z'y'$

$x'.y' + t' + z'\underbrace{(y'+1)}_{1}$

$$\boxed{f(x,y,z,t) = x'y' + t' + z'}$$

e) $f(x,y,z) = x'.y'.z + x'.y.z + x.y.z' + xyz + xyz'$

$x'(y'.z + y.z + y.z') + xy\underbrace{(z+z')}_{1}$

$x'(y'z + y(z+z')) + xy$

$x'(y'z + y) + xy$

$x'y'z + x'y + xy$

$x'y'z + y\underbrace{(x'+x)}_{1}$

$x'yz + y$

$(x'+y).\underbrace{(y+y')}_{1}.(z+y)$

$(x'+y).(z+y)$

$$\boxed{f(x,y,z) = x'z + y}$$

f) $f(x,y) = x'.y + x.y' + xy$

$= x'.y + x\underbrace{(y'+y)}_{1}$

$= x'y + x$

$= \underbrace{(x+x')}_{1}.(x+y)$

$\boxed{f(x,y) = x + y}$

g) $f(x,y,z) = [(x'.y'.z').(x+y+z')]'$

$[\underbrace{(x'.y'.z'.x)}_{0} + \underbrace{(y.x'.y'.z')}_{0} + (x'.y'.z'.z)]'$

$(x'.y'.z')' = x+y+z$

$\boxed{f(x,y,z) = x + y + z}$

h) $f(x,y,z) = x'.(x+y) + z' + zy$

$= \underbrace{x'.x}_{0} + x'y + z' + zy$

$= x'y + (zy + z')$

$= x'y + \underbrace{(z'+z)}_{1}.(z'+y)$

$= x'y + (z'+y)$

$= y\underbrace{(x'+1)}_{1} + z' + x'y$

$\boxed{F(x,y,z) = y + z' + x'y}$

(04)

i) $f(x,y) = (x+y'+xy).(x+y').(x'.y)$

$= [x\underbrace{(y+1)}_{1} + y'].(x+y').(x'y)$

$= (x+y').(x+y').(x'y)$

$= (x+y').(x'y)$

$= \underbrace{(x'yx)}_{0} + \underbrace{(x'yy')}_{0}$

$\boxed{f(x,y) = 0}$

ii) $f(x,y,z) = (x+y'+x.y).(xy+x'z+yz)$

$= [x\underbrace{(y'+1)}_{1} + y'].(xy+x'z+yz)$

$= (x+y').(xy+x'z+yz)$

$= (x+y')(xy) + (x+y').(x'z) + (x+y').(yz)$

$= (xy + \underbrace{xyy'}_{0}) + (\underbrace{x'zx}_{0} + y'x'z) + (xyz + \underbrace{y'yz}_{0})$

$= xy + x'y'z + xyz$

$= xy\underbrace{(z+1)}_{1} + x'y'z$

$\boxed{f(x,y,z) = xy + x'y'z}$

k) $f(x,y,z,t,u) = (xy+z+t).(z+t').(z+t'+u)$

$= (xy+z+t).((z+t').z + (z+t').t' + (z+t').u)$

$= (xy+z+t).(z+zt'+zt'+t'+zu+t'u)$

$= (xy+z+t).(z+zt'+t'+t'u+zu)$

$= (xy+z+t).(z\underbrace{(1+t')}_{1} + t'\underbrace{(1+u)}_{1} + zu)$

$= (xy+z+t).(z+t'+zu)$

$= (xy+z+t).(z\underbrace{(1+u)}_{1}+t')$

$= (xy+z+t).(z+t')$

$= (z+t').(xy) + z(z+t') + (z+t').t$

$= xyz + xyt' + z + zt' + z + 0$

$= xyz + xyt' + z\underbrace{(1+t')}_{1}$

$= xyz + xyt' + z$

$= z\underbrace{(xy+1)}_{1} + xyt'$

$\boxed{f(x,y,z,t,u) = xyt' + z}$

(a)

l) $f(x,y,z,t) = x' \cdot y \cdot (t' + tz') + (x + x'zt) \cdot y$

$= x'yt' + x'ytz' + yx + yx'zt$

$= x'(yt' + ytz' + yzt) + yx$

$= x'(yt' + yt\underbrace{(z' + z)}_{1}) + yx$

$= x'(y\underbrace{(t' + t)}_{1}) + yx$

$= x'y + yx$

$= y\underbrace{(x' + x)}_{1} \Rightarrow$ $\boxed{f(x,y,z,t) = y}$

m) $f(x,y,z,t) = (t + x + y) \cdot (t + x' + y) \cdot (y' + z) \cdot (t + z)$

$= (t + x + y) \cdot t + (t + x + y) \cdot x' + (t + x + y) \cdot y) \cdot (y' + z) \cdot (t + z)$

$t(t + x' + y)$

$= (t + tx + ty + \underbrace{tx + tx'}_{t} + \underbrace{x'y}_{y} + ty + xy + y) \cdot (y' + z) \cdot (t + z)$

$= (t + tx + ty + y) \cdot (y' + z) \cdot (t + z)$

$= (t\underbrace{(1 + x)}_{1} + y\underbrace{(t + 1)}_{1}) \cdot (y' + z) \cdot (t + z)$

$= (t + y) \cdot (y' + z) \cdot (t + z)$

$= (t + y)y' + (t + y)z) \cdot (t + z)$

$= (ty' + tz + zy) \cdot (t + z)$

$= (t + z)ty' + tz(t + z) + zy(t + z)$

$= ty' + tzy' + tz + tz + tzy + zy$

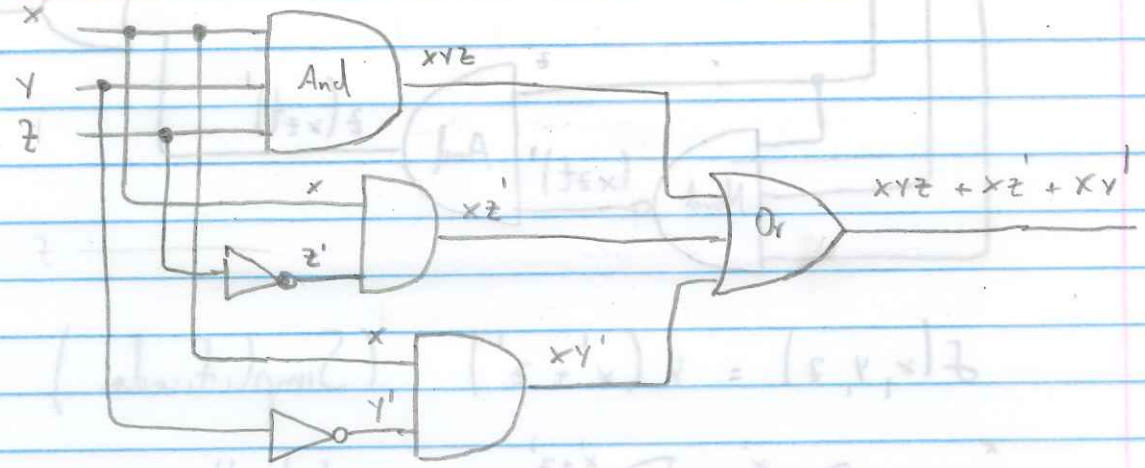
$= ty' + tz(y' + 1) + zy(t + 1)$

$\boxed{f(x,y,z,t) = ty' + tz + zy}$

Representação Gráfica

(Questão 1)

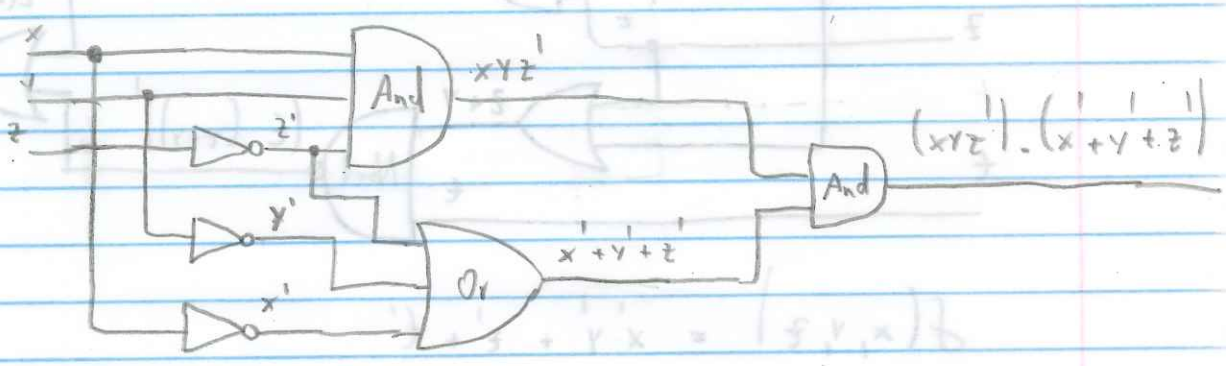
a) $f(x, y, z) = xyz + xz' + xy'$ (Original)



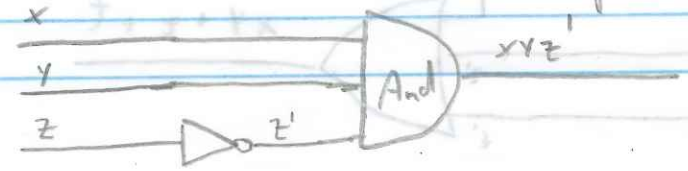
$f(x, y, z) = x$ (Simplificado)



b) $f(x, y, z) = (xyz')\,.\,(x' + y' + z')$ (original)

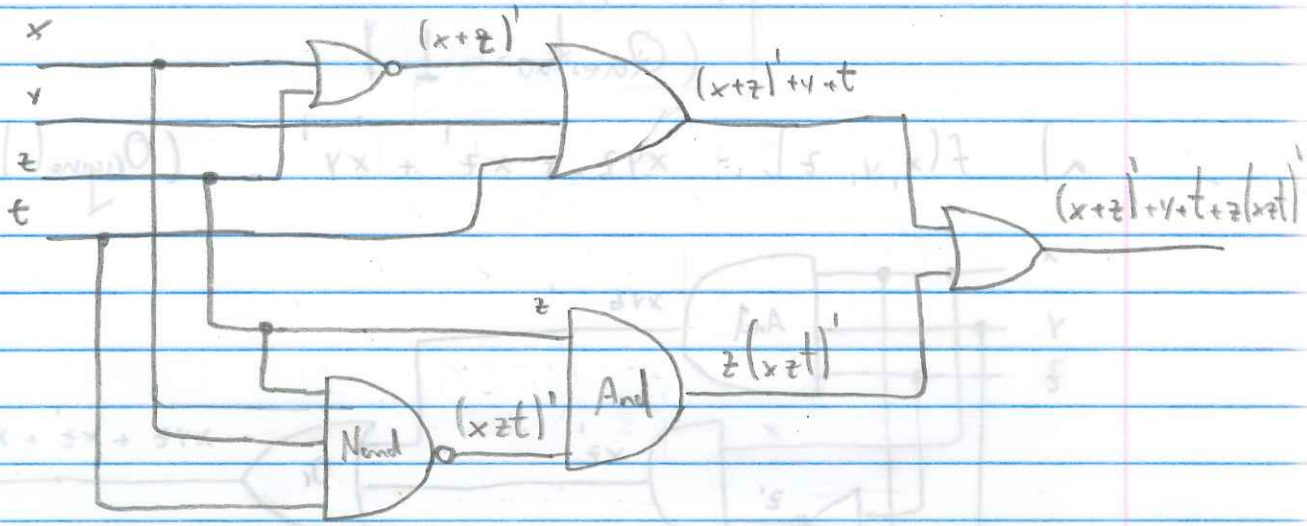


$f(x, y, z) = xyz'$ (Simplificado)

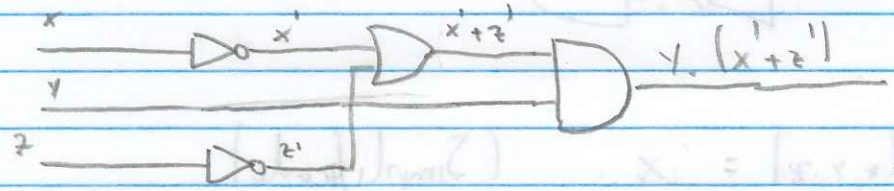


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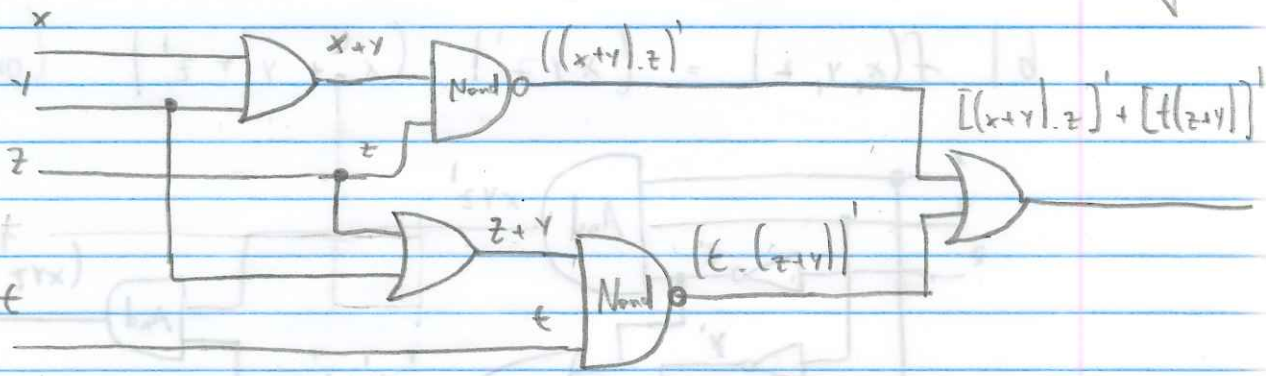
c) $f(x, y, z, t) = ((x+z)' + y + t) + z(xzt)'$ (Original)



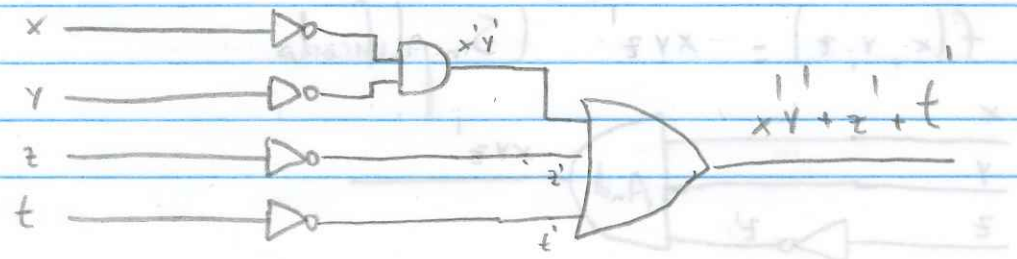
$f(x, y, z) = y.(x' + z')$ (Simplificada)



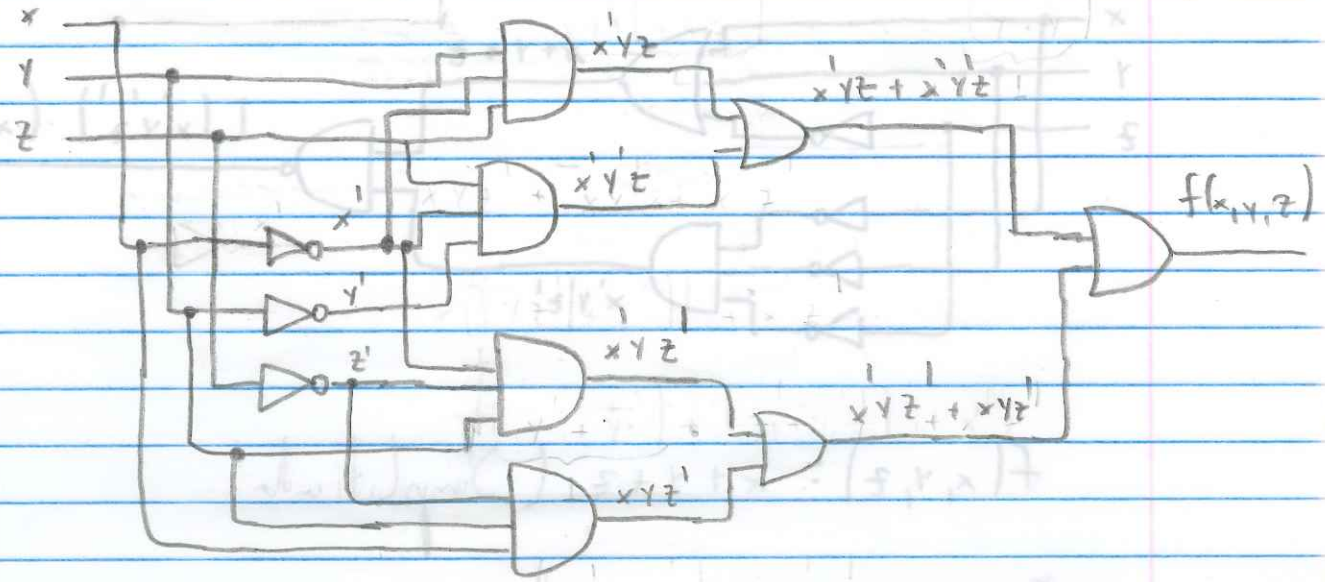
d) $f(x, y, z, t) = [(x+y).z]' + [t(z+y)]'$ (Original)



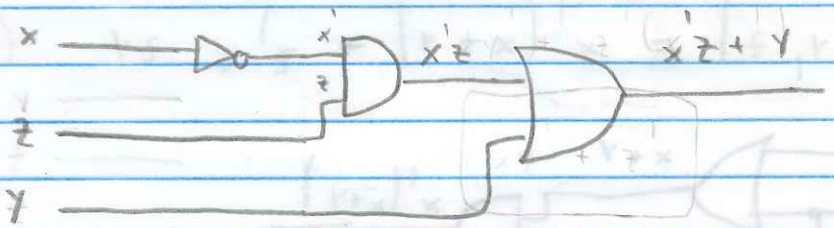
$f(x, y, z) = x'y' + z' + t'$



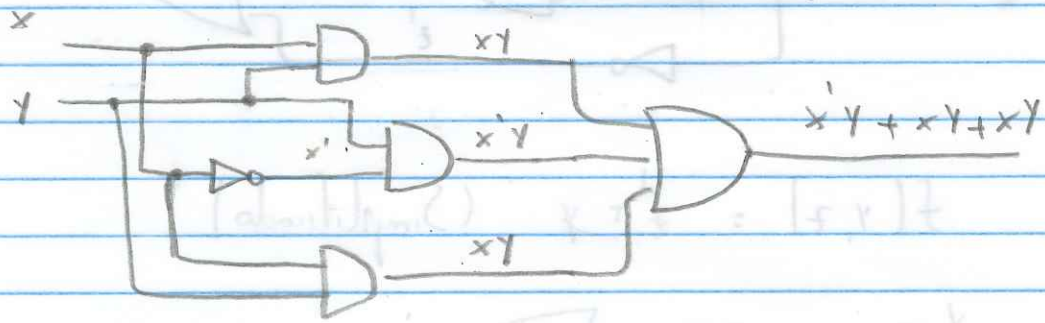
e) $f(x,y,z) = x'y'z + xy'z + xyz' + xyz$ (Original)



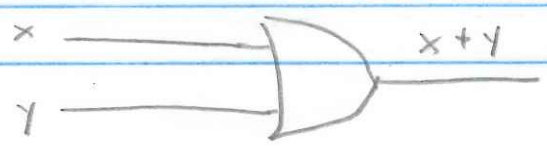
$f(x,y,z) = x'z + y$ (Simplificada)



f) $f(x,y) = x'y + xy + xy$ (Original)

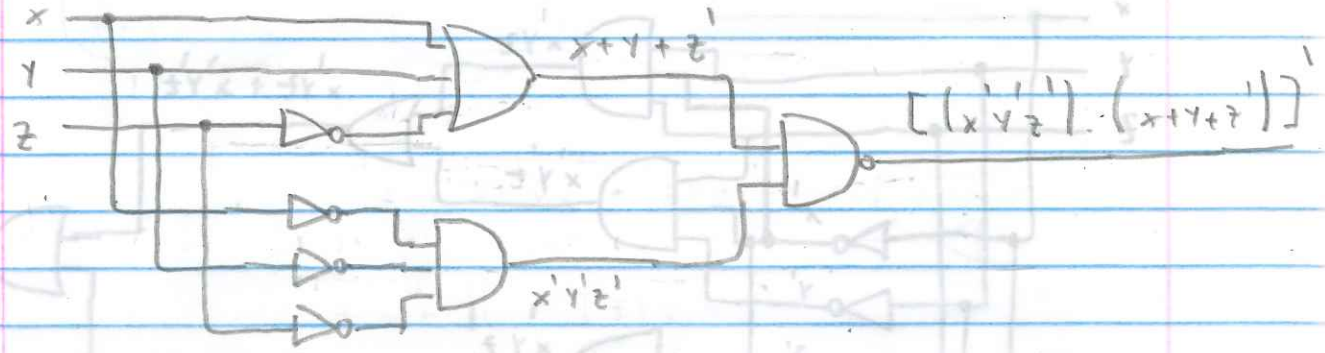


$f(x,y) = x + y$ (Simplificada)

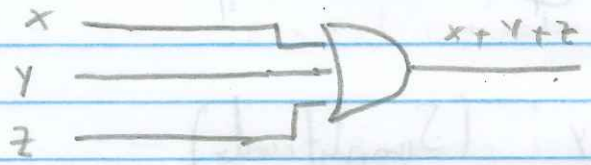


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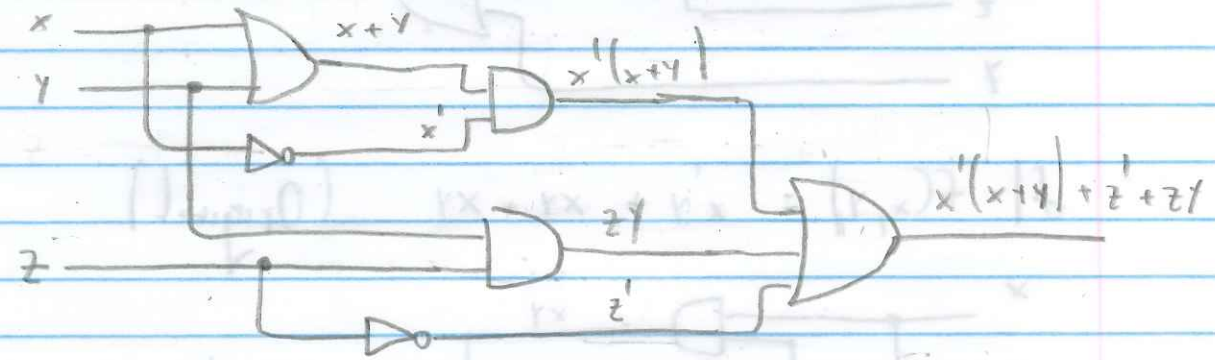
g) $f(x,y,z) = [x' \cdot y' \cdot z' \cdot (x + y + z')]'$ (Original)



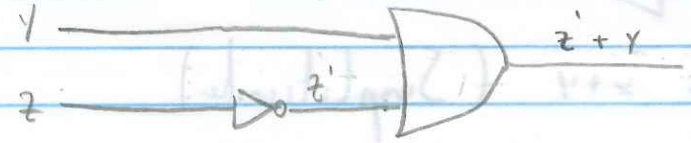
$f(x,y,z) = x + y + z$ (Simplificada)



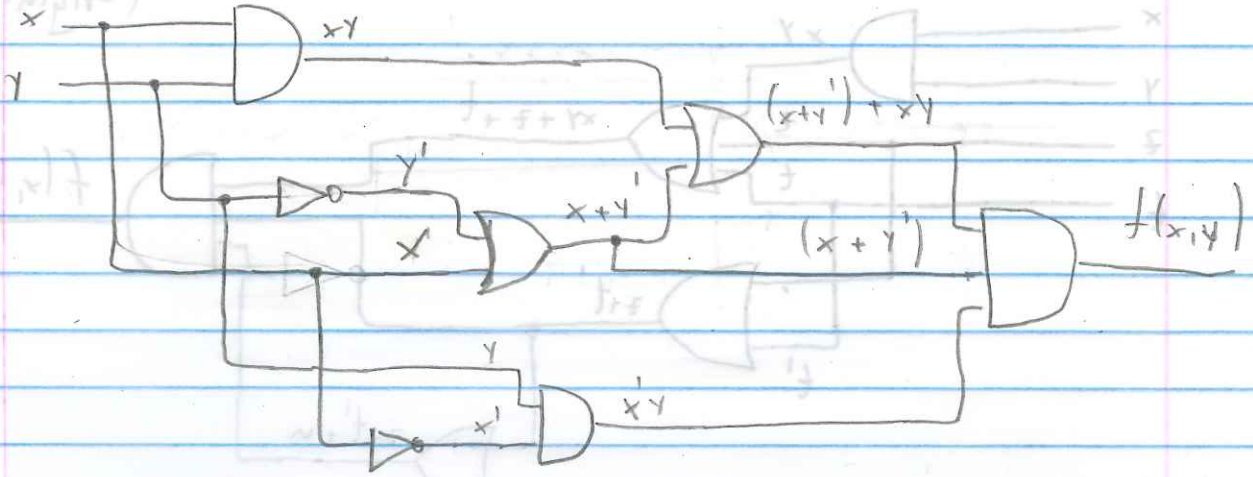
h) $f(x,y,z) = x'(x + y) + z' + zy$ (Original)



$f(y,z) = z' + y$ (Simplificado)

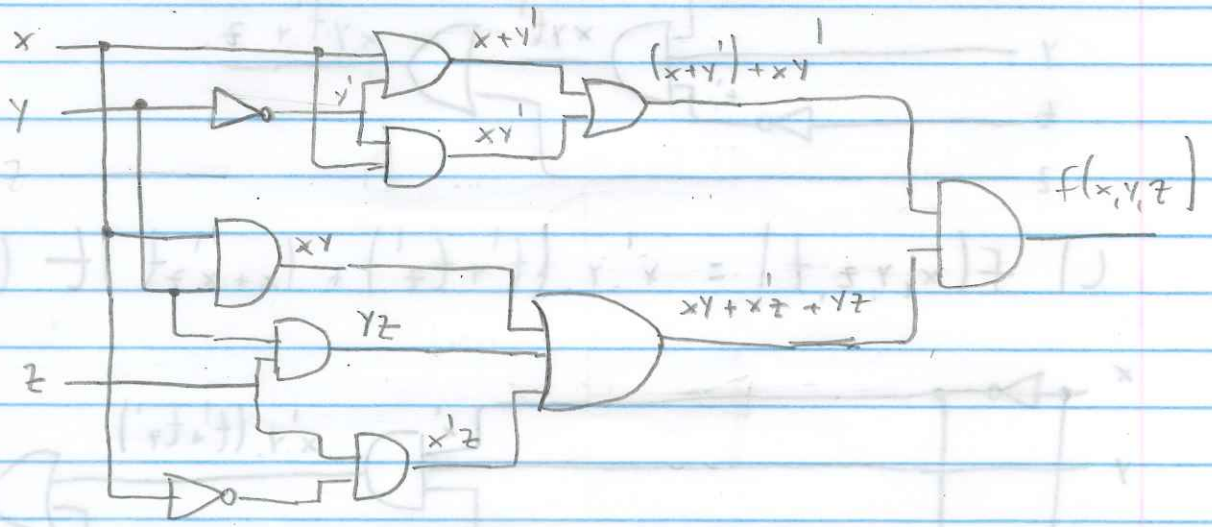


i) $f(x,y) = (x + y' + xy) \cdot (x + y') \cdot (x' + y)$ (Original)

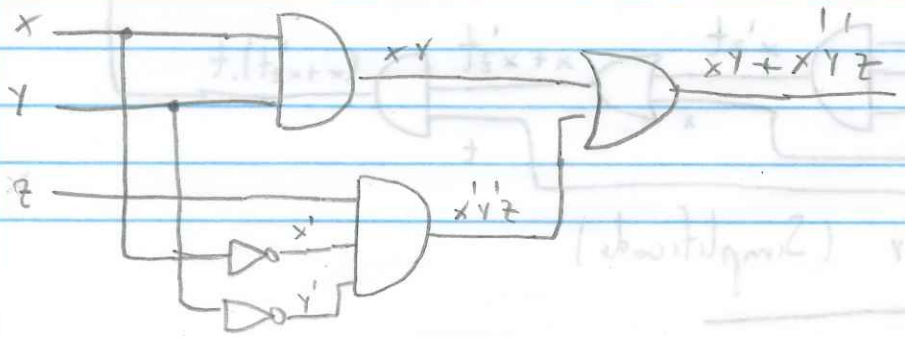


$f(x,y) = 0$ (Simplificada) ✗

ii) $f(x,y,z) = (x + y' + xy') \cdot (xy + x'z + yz)$ (Original)

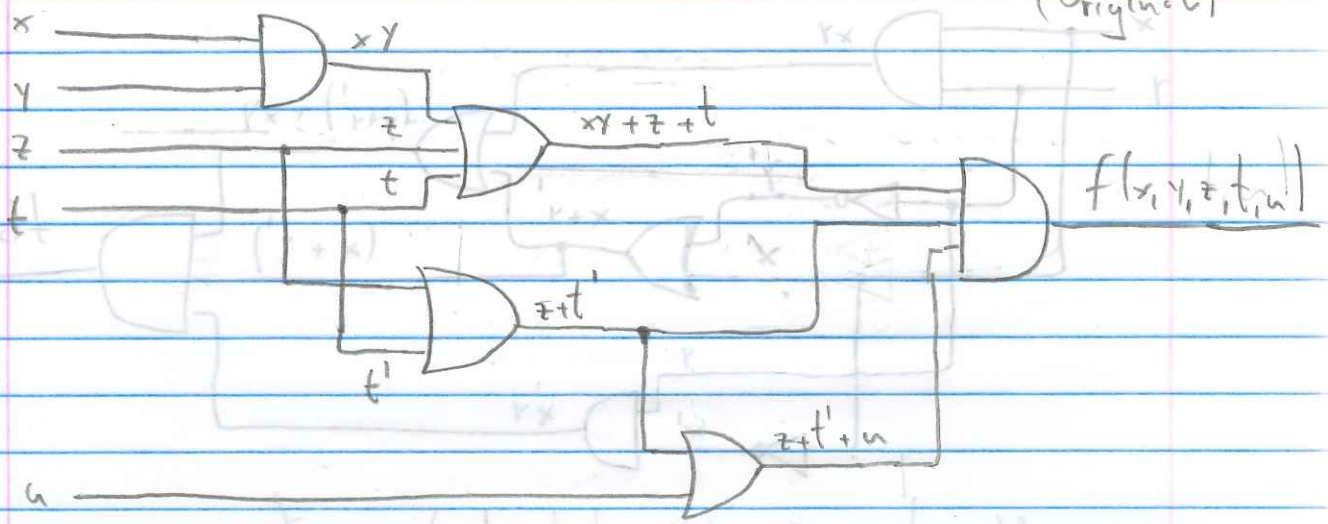


$f(x,y,z) = xy + x'y'z$

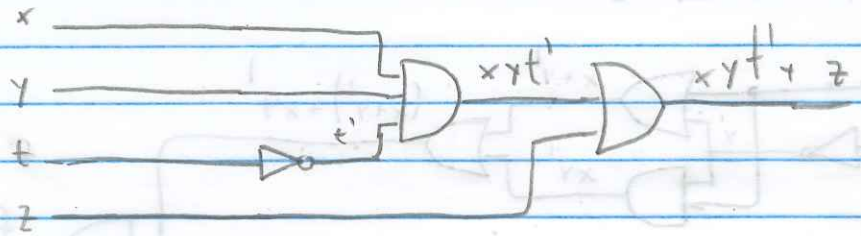


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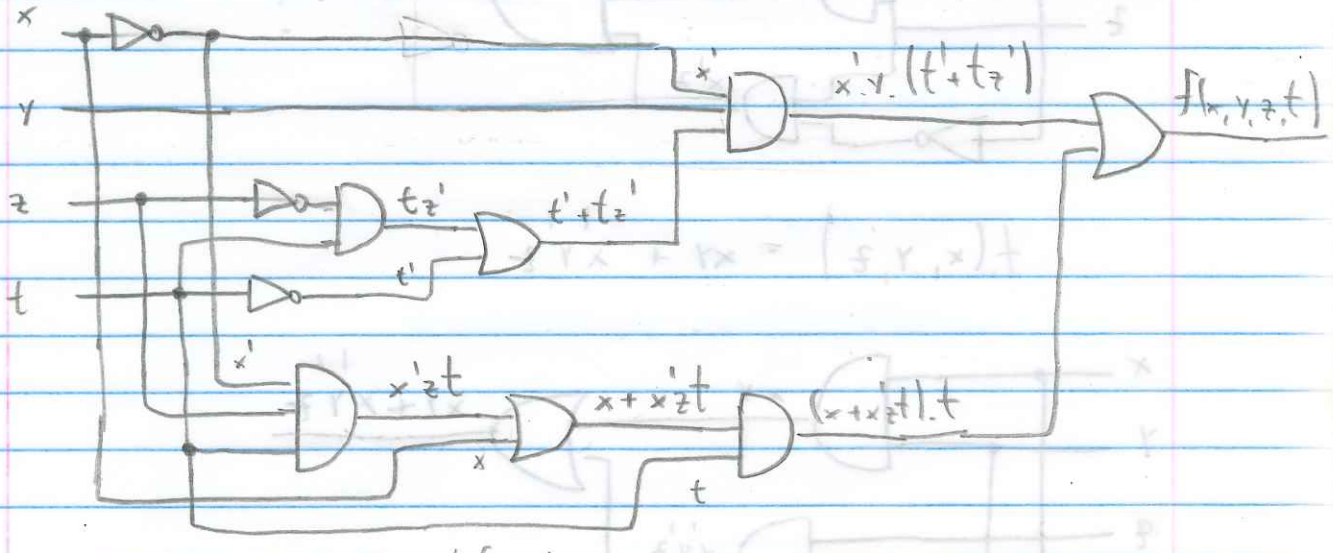
k) $f(x,y,z,t,u) = (xy + z + t).(z + t').(z + t' + u)$ (Original)



$f(x,y,z) = xyt' + z$ (Simplificada)



l) $f(x,y,z,t) = x'.y.(t' + tz') + (x + x'zt).t$ (Original)

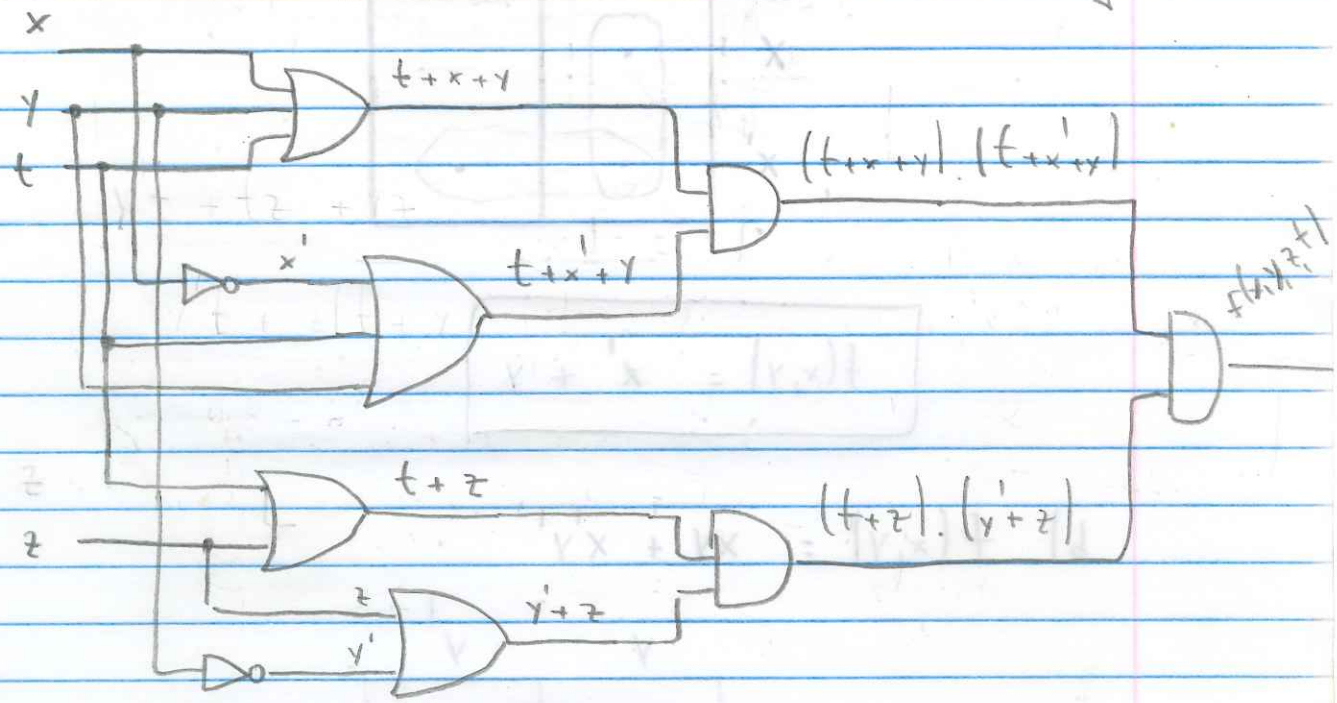


$f(y) = y$ (Simplificada)

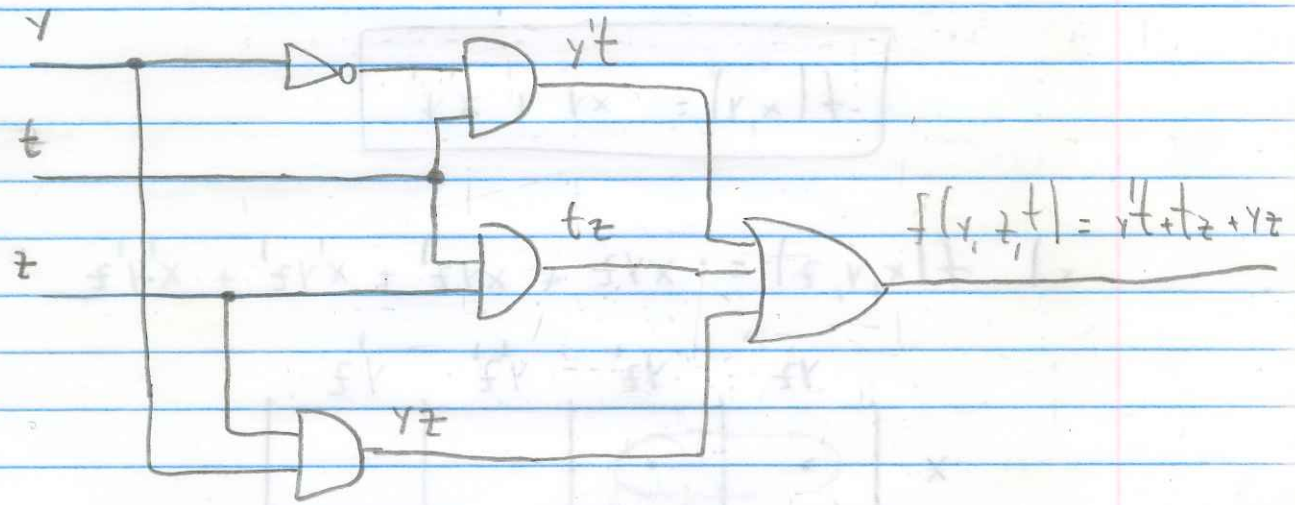
y

m) $f(x, y, z, t) = (t + x + y)\,(t + x' + y)\,(y' + z)\,(t + z)$

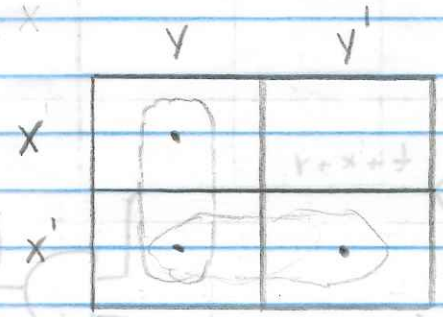
(Original)



$$f(x, z, t) = y't + tz + yz$$

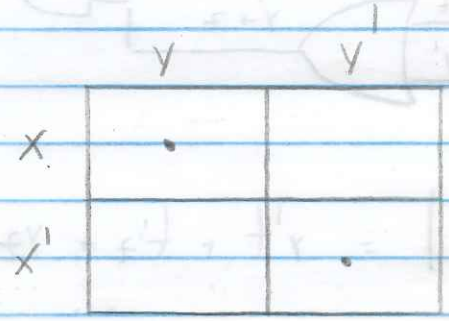


(02) a) $f(x,y) = xy + x'y + x'y'$



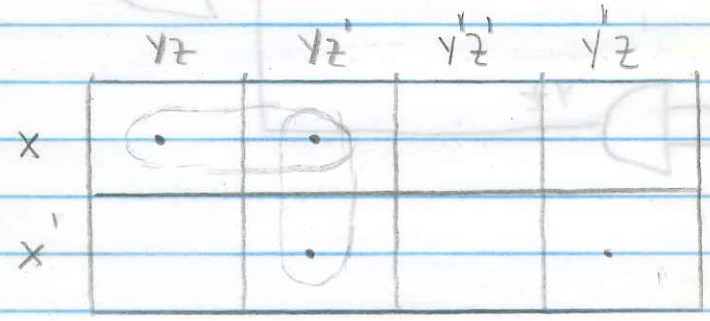
$$f(x,y) = x' + y$$

b) $f(x,y) = xy + x'y'$



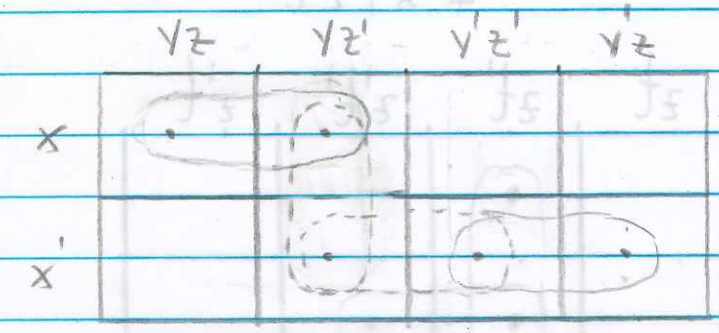
$$f(x,y) = xy + x'y'$$

c) $f(x,y,z) = xyz + xyz' + x'yz' + x'y'z$



$$f(x,y,z) = xy + yz' + x'y'z$$

d) $f(x,y,z) = xyz + xyz' + xy'z' + x'yz' + x'y'z$



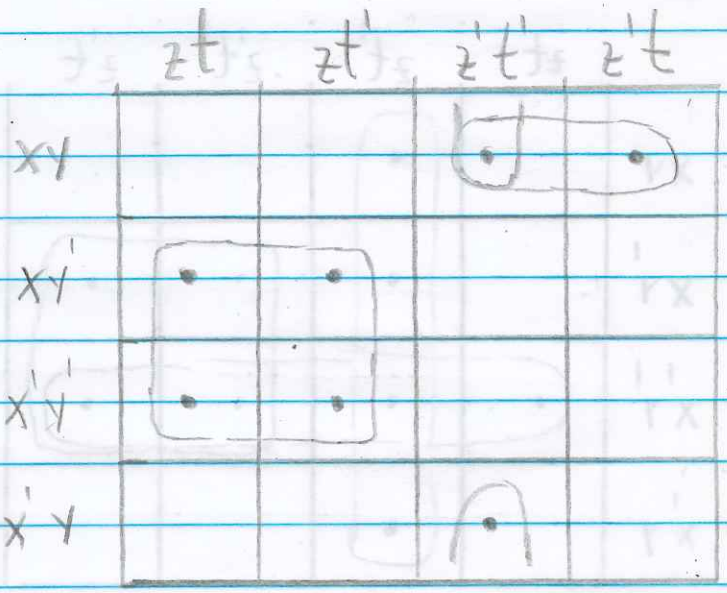
$$f(x,y) = xy + yz' + x'y'$$

ou

$$f(x,y) = xy + xz' + x'y'$$

("cobertura mínima")

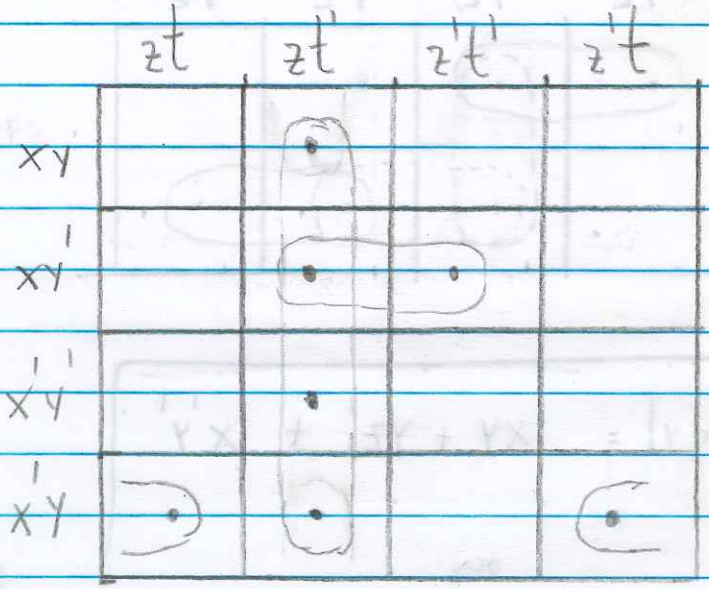
e) $f(x,y,z,t) = xyzt' + xyz't + xyz't' + xy'zt + xy'z't' + x'y'zt + x'y'z't' + x'yz't'$



$$f(x,y,z,t) = y'z + xyz' + yz't'$$

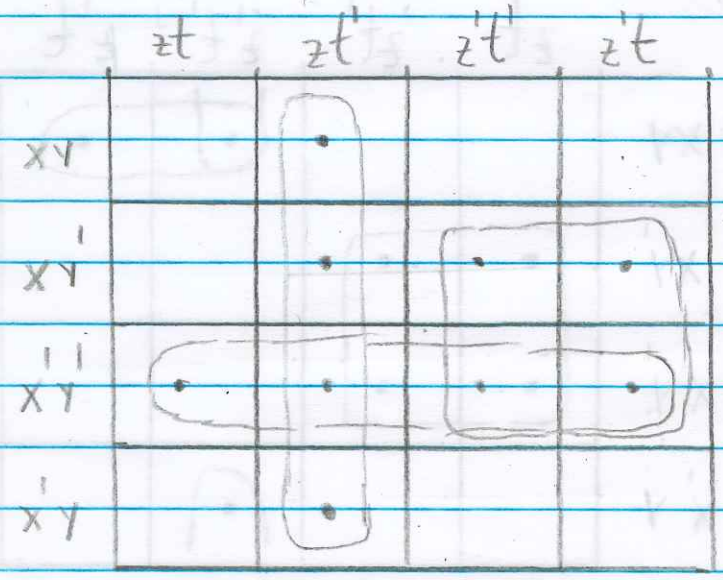
16

f) $f(x,y,z,t) = x'yz + x'yz't + y'zt' + xyzt' + xy'z't'$



$$f(x,y,z,t) = zt' + xy't' + x'yt$$

g) $f(x,y,z,t) = y't' + y'z't + x'y'zt + yzt'$

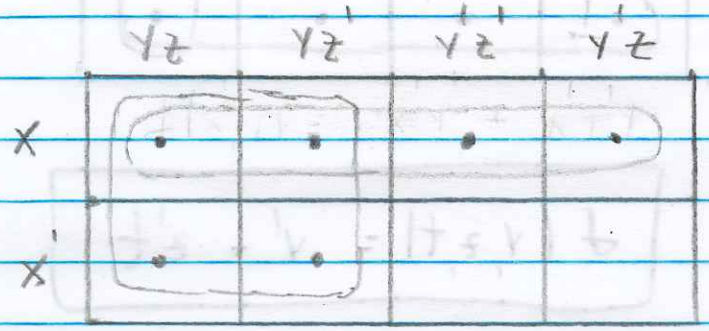


$$f(x,y,z,t) = x'y' + zt' + y'z'$$

h) $f(x,y,z) = (x+y)(xz + xz') + xy + y$

Devemos transformar em uma soma de produtos

$$f(x,y,z) = xz + xz' + yxz + yxz' + xy + y$$

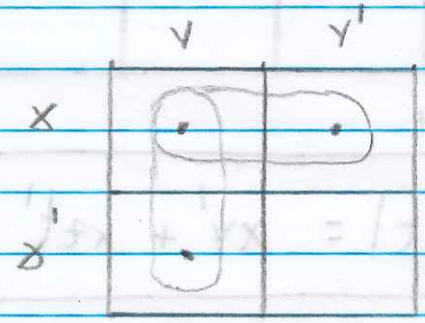


$$\boxed{f(x,y) = x + y}$$

i) $f(x,y) = x'(x+y) + (y+x)(x+y')$

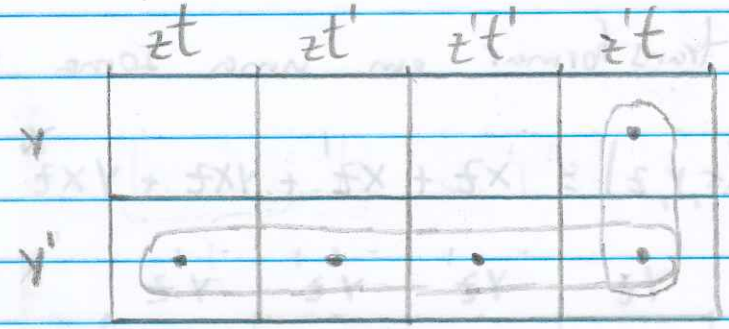
Transformando em uma soma de produtos

$$f(x,y) = x'y + xy + x + xy'$$



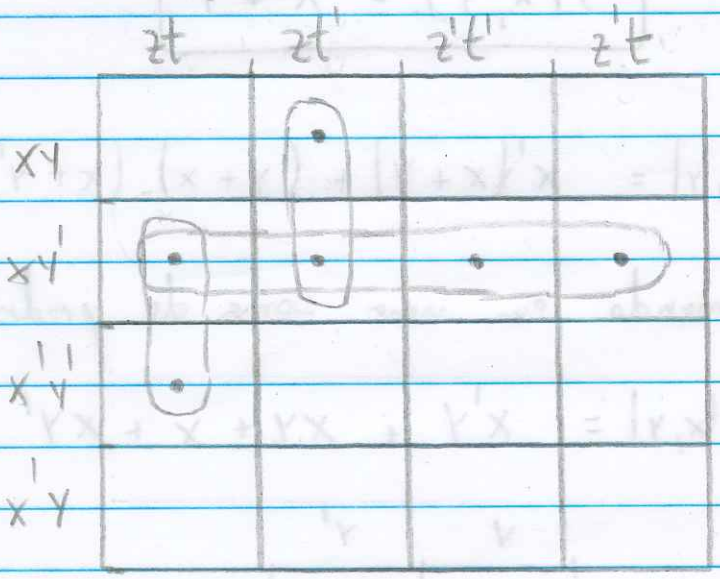
$$\boxed{f(x,y) = x + y}$$

3) $f(y, z, t) = y'z + y'z't' + z't$



$$f(y, z, t) = y' + z't$$

4) $f(x, y, z, t) = y'zt + xzt' + xy'z'$



$$f(x, y, z, t) = xy' + xzt' + y'zt$$